

No. Parallel Environments

VS

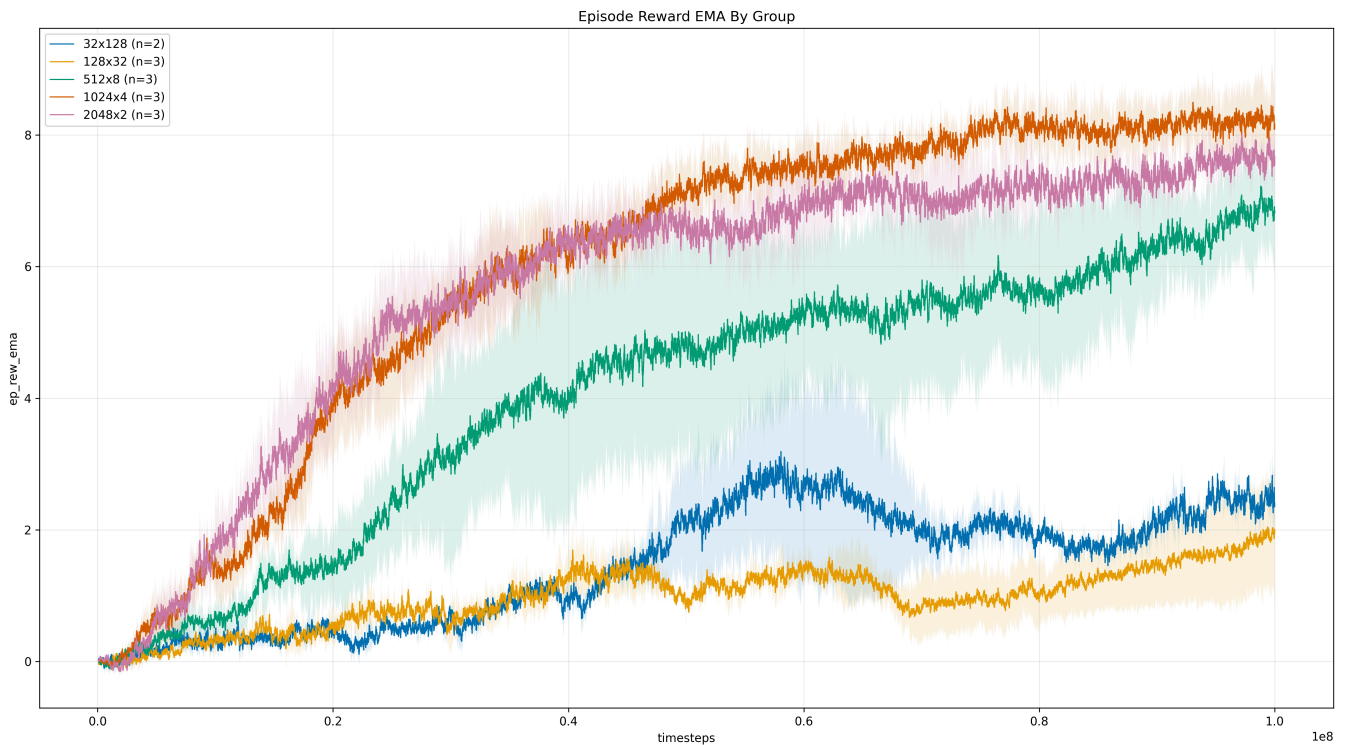
Episode Segment Length

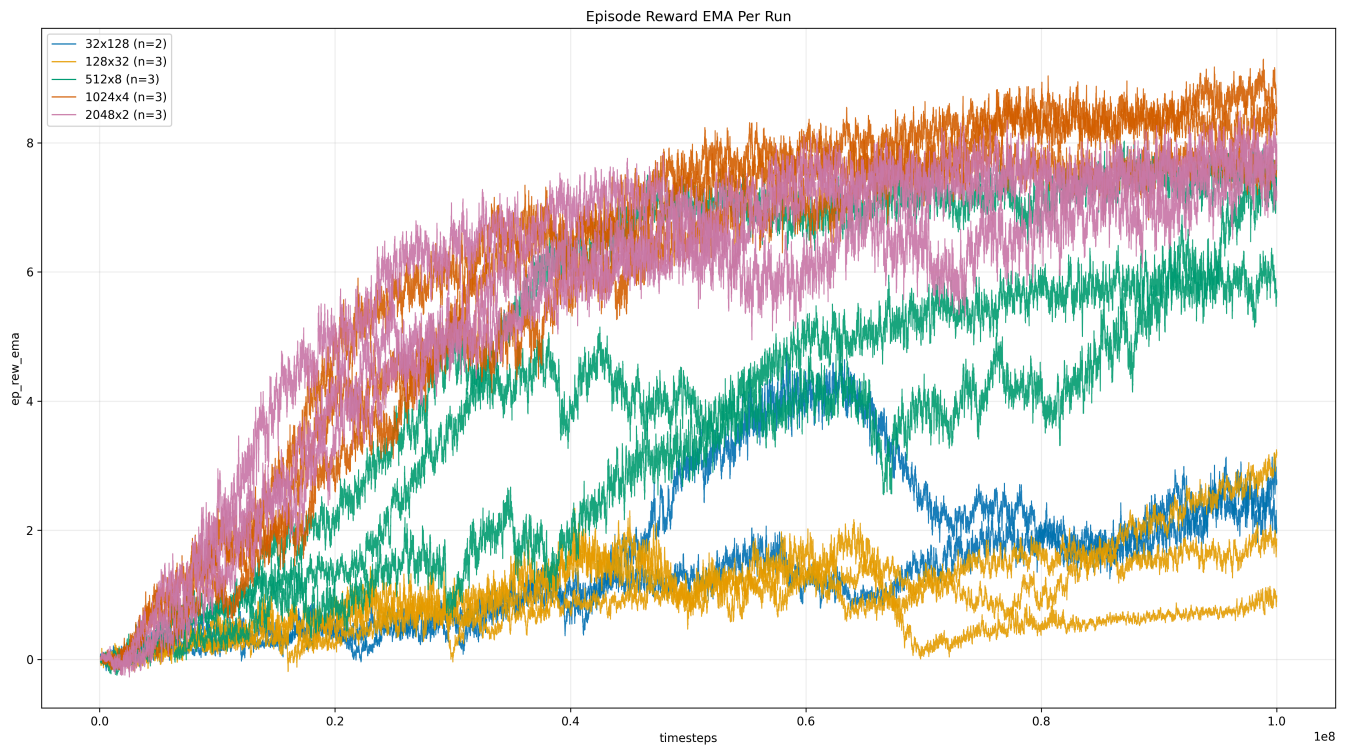
This experiment sets out to test if MAT on the swarm-bots wall task benefits more from batches made up of a large amount of parallel environments or of long episode segments. We fix the batch size $B = 4096$ and iterate through combinations of parallel environments P and episode segment lengths T where $B = P \times T$. Concretely, the following combinations were used:

- 32×128
- 128×32
- 512×8
- 1024×4
- 2048×2

For the last combination it's important to note that the next-observation-prediction auxiliary loss normally predicts up to 4 steps into the future. Since we have $T = 2$, we can only predict up to two steps.

Results





Although we do not have many runs per category, a clear trend is visible. Higher P leads to more stable and faster learning. 2048×2 learns slightly faster in the beginning compared to 1024×4 but ends up with a lower final return.

Interpretation

While long episode segment lengths increase the accuracy of advantage and value estimates, MAT in the swarm-bots wall scenario appears to favor the diversity that comes from having many parallel environments. Having many environments also means fewer steps in the batch are correlated with each other. This likely reduces batch bias and thus leads to more stable learning.

Next Steps

This experiment suggests that MAT favors batches with high diversity. The obvious next step would be to increase the batch size B to make room for higher P e.g 2048×4 . However, [MAPPO](#) has shown that using excessively large batch sizes might reduce sample efficiency, so we will have to test where the ceiling will be.